

Matrices and Systems of Linear Equations (2005–2014)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

8

TOTAL MARKS

40 M

TEST DURATION

24 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		8	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q12 • ID: JAM_2014_Q12)

MCQ

+2 M

Neg: -0.67

- Q.12 Let $A = \begin{bmatrix} a & -1 & 4 \\ 0 & b & 7 \\ 0 & 0 & 3 \end{bmatrix}$ be a matrix with real entries. If the sum and the product of all the eigenvalues of A are 10 and 30 respectively, then $a^2 + b^2$ equals
- (A) 29 (B) 40 (C) 58 (D) 65

Question 2 (JAM 2014 • Q11 • ID: JAM_2014_Q11)

MCQ

+2 M

Neg: -0.67

- Q.11 The system of linear equations

$$x - y + 2z = b_1$$

$$x + 2y - z = b_2$$

$$2y - 2z = b_3$$

is inconsistent when (b_1, b_2, b_3) equals

- (A) (2, 2, 0) (B) (0, 3, 2) (C) (2, 2, 1) (D) (2, -1, -2)

Question 3 (JAM 2012 • Q12 • ID: JAM_2012_Q12)

MCQ

+6 M

Neg: -2.0

- Q.12 Let P be a 4×4 matrix whose determinant is 10. The determinant of the matrix $-3P$ is
- (A) -810 (B) -30 (C) 30 (D) 810

Question 4 (JAM 2011 • Q11 • ID: JAM_2011_Q11)

MCQ

+6 M

Neg: -2.0

- Q.11 Let A be a 3×3 matrix with $\text{trace}(A) = 3$ and $\det(A) = 2$. If 1 is an eigenvalue of A , then the eigenvalues of the matrix $A^2 - 2I$ are

(A) 1, $2(i-1)$, $-2(i+1)$

(B) -1, $2(i-1)$, $2(i+1)$

(C) 1, $2(i+1)$, $-2(i+1)$

(D) -1, $2(i-1)$, $-2(i+1)$

Question 5 (JAM 2008 • Q2 • ID: JAM_2008_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 Let $S = \{T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \mid T \text{ is a linear transformation with } T(1,0,1) = (1,2,3), T(1,2,3) = (1,0,1)\}$. Then S is

- (A) a singleton set
(C) a countably infinite set

- (B) a finite set containing more than one element
(D) an uncountable set

Question 6 (JAM 2007 • Q11 • ID: JAM_2007_Q11)

MCQ

+6 M

Neg: -2.0

11. The rank of the matrix $\begin{bmatrix} 1 & 4 & 8 \\ 2 & 10 & 22 \\ 0 & 4 & 12 \end{bmatrix}$ is

- (A) 3
(B) 2
(C) 1
(D) 0

Question 7 (JAM 2006 • Q11 • ID: JAM_2006_Q11)

MCQ

+6 M

Neg: -2.0

11. If A and B are 3×3 real matrices such that $\text{rank}(AB) = 1$, then $\text{rank}(BA)$ **cannot** be

- (A) 0
(B) 1
(C) 2
(D) 3

Question 8 (JAM 2005 • Q6 • ID: JAM_2005_Q6)

MCQ

+6 M

Neg: -2.0

6. Let A be a 3×3 matrix with eigenvalues 1, -1 and 3. Then

- (A) $A^2 + A$ is non-singular
(B) $A^2 - A$ is non-singular
(C) $A^2 + 3A$ is non-singular
(D) $A^2 - 3A$ is non-singular

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Matrices and Systems of Linear Equations (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	A
2	MCQ	+2	C
3	MCQ	+6	D
4	MCQ	+6	D
5	MCQ	+6	D
6	MCQ	+6	B
7	MCQ	+6	D
8	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.